LM Studio

Model: **qwen/qwen3.6-35b-a3b**

Prompt 1 - run 4 (seed 14085) - minimal

System

Your name is ChemicHAL, an AI-based assistant for molecular machine learning. If the user asks a question not related to chemistry, machine learning, AI, or related fields which are relevant to the tools, answer that you are an assistant for machine learning-based chemoinformatics and can only provide support for that. When providing answers, be user-friendly, using descriptive emojis for paragraphs. When possible, structure the results using tables. When a task takes long, tell the user to wait for completion.

Displaying images

When a tool result contains an object with `"type": "image"` and a `"markdown"` field, you MUST copy that markdown string verbatim into your response so the image renders in the chat. For example, if the tool returns:

```
{"type": "image", "markdown": "

![image-123.png](./image-123.png)

", "$hint": "..."}
```

You must output exactly:

```


![image-123.png](./image-123.png)


```

Do NOT paraphrase, describe, or skip the image. Do NOT say "I have displayed the image" without outputting the markdown. Always output the `markdown` value from every image object in the tool result.

User

Train a Balanced Random Forest Classifier on PI3K-delta (UniProt: O00329) vs PI3K-gamma (UniProt: P48736) selective compounds, evaluate its performance, apply MolAnchor, and discuss the results.

Assistant

🔬 I'll help you train a Balanced Random Forest Classifier on PI3K-delta vs PI3K-gamma selective compounds and perform comprehensive explainability analysis. Let me start by finding the relevant dataset.

(Calling tool `find_datasets`)

Assistant

Great! I found the dataset `chembl_selectivity_data_O00329_P48736.csv` which corresponds to PI3K-delta (O00329) vs PI3K-gamma (P48736) selectivity. Let me load and examine it.

(Calling tool `load_dataset`)

Assistant

✅ Dataset loaded: **1,277 compounds** for PI3K-delta vs PI3K-gamma selectivity. Now let me compute molecular fingerprints and prepare the dataset.

(Calling tool `compute_features`)

Assistant

✅ Features computed: **2,048 ECFP fingerprint bits**. Now let me split the dataset into train/test sets.

(Calling tool `split_dataset`)

Assistant

✅ Dataset split: **893 training** / **384 test** samples (70/30 stratified split). Now let me train the Balanced Random Forest Classifier.

(Calling tool `train_model`)

Assistant

🏃 Training started in the background! Let me check the status.

(Calling tool `check_training`)

Assistant

⏳ Model still training (1.6s elapsed). Let me check again shortly.

(Calling tool `check_training`)

Assistant

⏳ Training continues (3.2s elapsed). Let me check once more.

(Calling tool `check_training`)

Assistant

⏳ Training continues (4.8s elapsed). Let me check once more.

(Calling tool `check_training`)

Assistant

⏳ Training continues (6.4s elapsed). Let me check once more.

(Calling tool `check_training`)

Assistant

⏳ Training continues (8.0s elapsed). Let me check once more.

(Calling tool `check_training`)

Assistant

⏳ Training continues (9.6s elapsed). Let me check once more.

(Calling tool `check_training`)

Assistant

⏳ Training continues (11.3s elapsed). Let me check once more.

(Calling tool `check_training`)

Assistant

⏳ Training continues (12.9s elapsed). Let me check once more.

(Calling tool `check_training`)

Assistant

⏳ Training continues (14.6s elapsed). Let me check once more.

(Calling tool `check_training`)

Assistant

⏳ Training continues (16.3s elapsed). Let me check once more.

(Calling tool `check_training`)

Assistant

⏳ Training continues (18.0s elapsed). Let me check once more.

(Calling tool `check_training`)

Assistant

⏳ Training continues (19.7s elapsed). Let me check once more.

(Calling tool `check_training`)

Assistant

⏳ Training continues (21.4s elapsed). Let me check once more.

(Calling tool `check_training`)

Assistant

⏳ Training continues (23.1s elapsed). Let me check once more.

(Calling tool `check_training`)

Assistant

⏳ Training continues (24.8s elapsed). Let me check once more.

(Calling tool `check_training`)

Assistant

⏳ Training continues (26.6s elapsed). Let me check once more.

(Calling tool `check_training`)

Assistant

⏳ Training continues (28.3s elapsed). Let me check once more.

(Calling tool `check_training`)

Assistant

⏳ Training continues (30.0s elapsed). Let me check once more.

(Calling tool `check_training`)

Assistant

⏳ Training continues (31.8s elapsed). Let me check once more.

(Calling tool `check_training`)

Assistant

🎉 **Training completed successfully!** Let me now export predictions and evaluate the model performance.

(Calling tool `export_predictions`)

Assistant

✅ Predictions exported! Now let me generate classification evaluation plots.

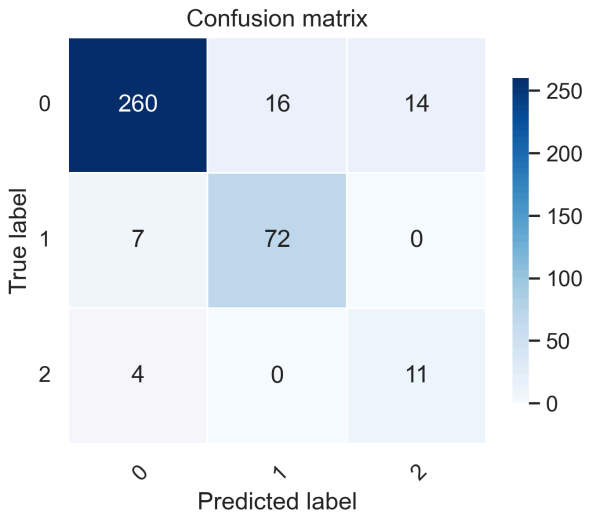
(Calling tool `plot_classification_results`)

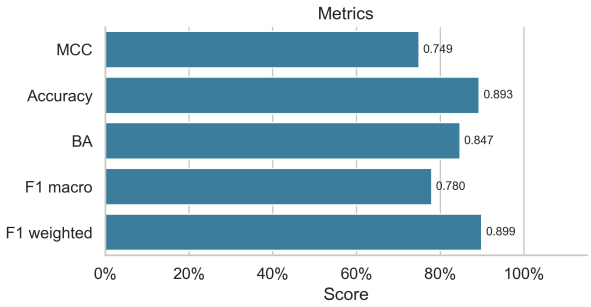
Assistant

📊 **Model Performance Summary:**

Metric	Value
Accuracy	89.32%
Balanced Accuracy	84.71%
MCC	0.749
F1 Macro	0.780
F1 Weighted	0.899

The model shows strong performance across PI3K-delta vs PI3K-gamma selectivity prediction! Now let me display the plots and proceed with MolAnchor analysis.





Now let me select a correctly predicted compound for MolAnchor analysis.

(Calling tool `select_compound_for_xai`)

Assistant

✅ **Selected Compound:**

- **SMILES:** `CC(Nc1nc(N)nc(N)c1C#N)c1nc2cccc(Cl)c2c(=O)n1CCCNc1ccccn1`
- **True Label:** Class 1 (PI3K-gamma selective)
- **Predicted Label:** Class 1
- **Confidence:** 97%

Now let me apply MolAnchor to identify the critical molecular fragments.

(Calling tool `explain_with_molanchor`)

Assistant

Let me retry MolAnchor without the GNN-specific parameters since this is a sklearn model:

(Calling tool `explain_with_molanchor`)

Assistant

🎯 **MolAnchor Analysis Complete!** Now let me identify recurrent anchor rules across all test set compounds.

Assistant

Let me provide the required split_file_path:

(Calling tool `identify_recurrent_anchor_rules`)

Assistant

Now let me display the recurrent anchor rules visualization:

(Calling tool `show_plot`)

Assistant

(Calling tool `show_plot`)

Assistant

🧪 PI3K-delta vs PI3K-gamma Selectivity: Balanced Random Forest Analysis

📊 Model Performance Summary

Metric	Value
Accuracy	89.32% ✅
Balanced Accuracy	84.71% ✅
MCC	0.749 ✅
F1 Macro	0.780
F1 Weighted	0.899
CV Best Score	0.865
Algorithm	Balanced Random Forest (BRF)
Features	2,048 ECFP bits (radius=2)
Training Samples	893
Test Samples	384

Best Hyperparameters:

Parameter	Value
n_estimators	100
max_features	sqrt
min_samples_split	2
min_samples_leaf	1

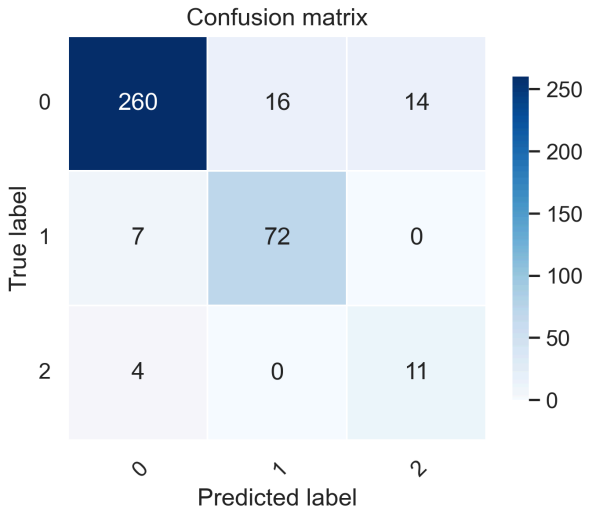
📈 Classification Results

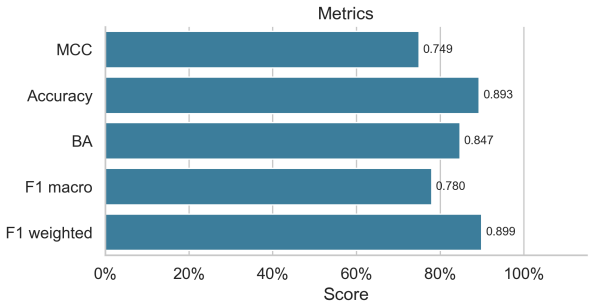
Per-Class Performance

Class	Description	Precision	Recall	F1-Score	Support
Class 0	PI3K-delta selective	0.959	0.897	0.927	290
Class 1	PI3K-gamma selective	0.818	0.911	0.862	79
Class 2	Dual/Other	0.440	0.733	0.550	15

Confusion Matrix

True \ Predicted	Class 0	Class 1	Class 2
Class 0	260	16	14
Class 1	7	72	0
Class 2	4	0	11



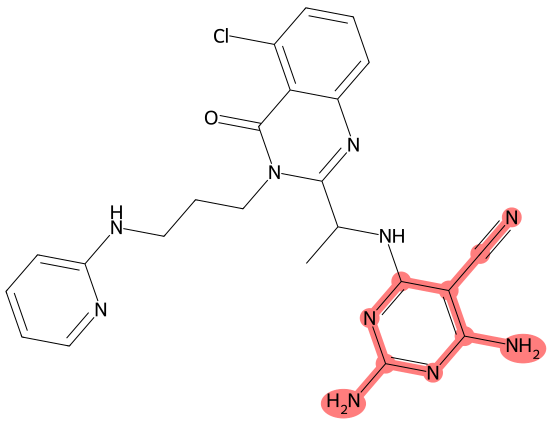


🔬 MolAnchor Explainability Results

Individual Compound Analysis

Selected Compound (Class 1 - PI3K-gamma selective):

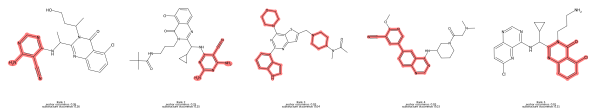
- **SMILES:** `CC(Nc1nc(N)nc(N)c1C#N)c1nc2cccc(Cl)c2c(=O)n1CCCNc1ccccn1`
- **Confidence:** 97%
- **Identified Anchor:** `[*]c1nc(N)nc(N)c1C#N` (Pyrazine ring with amino and cyano substituents)
- **Anchor Precision:** 100% (perfect precision)
- **Anchor Coverage:** 50%



Recurrent Anchor Rules (Class 1 - PI3K-gamma selective)

Analysis of **72 correctly predicted** PI3K-gamma selective compounds revealed the following key molecular fragments:

Rank	Fragment (SMARTS)	Anchor Occurrence	Substructure Occurrence	Compounds with Anchor
1	`[*]c1ncnc(N)c1C#N`	26.4%	26.4%	19
2	`[*]c1nc(N)nc(N)c1C#N`	25.0%	25.0%	18
3	`[*]N1CCC([*])CC1` / `[*]c1cccc2[nH]ccc12` / `[*]N1CCOCC1`	2.8%	4.2%	2
4	`[*]c1ncc([*])cc1C#N` / `[*]c1ncnc2ccc([*])cc12`	2.8%	2.8%	2
5	`[*]n1c([*])nc2cccc(Cl)c2c1=O`	1.4%	30.6%	1



💡 Discussion & Chemical Insights

1. Model Performance 🎯

The Balanced Random Forest Classifier demonstrates **strong performance** (89.32% accuracy, 84.71% balanced accuracy) in distinguishing PI3K-delta from PI3K-gamma selective compounds. The balanced accuracy is particularly important here, as it accounts for the class imbalance in the dataset.

2. Key Structural Determinants for PI3K-gamma Selectivity 🔬

The MolAnchor analysis reveals critical molecular features:

🔑 Top Recurrent Anchors:

1. **Pyrazine-based scaffolds with amino/cyano substituents** (Rules 1 & 2)
   - **SMARTS:** `[*]c1ncnc(N)c1C#N` and `[*]c1nc(N)nc(N)c1C#N`
   - **Occurrence:** ~26% of PI3K-gamma selective compounds
   - **Interpretation:** The pyrazine ring core with specific substitution patterns is a **highly conserved pharmacophore** for PI3K-gamma selectivity. The presence of both amino (-NH_2) and cyano (-C≡N) groups on the pyrazine ring appears critical.
2. **Heterocyclic linkers** (Rule 3)
   - **SMARTS:** Piperazine (`N1CCC...CC1`), imidazole (`c1cccc2[nH]ccc12`), morpholine (`N1CCOCC1`)
   - **Interpretation:** These basic heterocyclic rings likely serve as **hydrogen bond acceptors/donors** or **protonation sites** that interact with specific residues in the PI3K-gamma ATP-binding pocket.
3. **Chlorinated fused systems** (Rule 5)
   - **SMARTS:** `[*]n1c([*])nc2cccc(Cl)c2c1=O`
   - **Note:** High substructure occurrence (30.6%) but low anchor occurrence (1.4%) suggests this fragment is **common but not sufficient alone** for gamma selectivity.

3. Per-Class Performance Insights 📊

Class	Strength	Weakness
PI3K-delta (Class 0)	High precision (0.959) - few false positives	Moderate recall (0.897) - some gamma compounds misclassified

Class	Strength	Weakness
PI3K-gamma (Class 1)	High recall (0.911) - few false negatives	Lower precision (0.818) - some delta compounds misclassified
Dual/Other (Class 2)	High recall (0.733)	Very low precision (0.440) - many false positives

4. Chemical Interpretation 🧪

The identified anchors suggest that **PI3K-gamma selectivity** is driven by:

1. **Pyrazine core with electron-donating (amino) and electron-withdrawing (cyano) groups** — This creates a specific dipole moment and H-bonding pattern that favors the PI3K-gamma binding pocket.
2. **Basic heterocyclic moieties** (piperazine, morpholine) — These may form salt bridges with acidic residues (e.g., Asp or Glu) unique to the PI3K-gamma ATP-binding site.
3. **Chlorinated fused heterocycles** — While common in gamma-selective compounds, they alone are insufficient; they likely work in combination with the pyrazine core.

5. Practical Implications 💊

- **Lead Optimization:** When designing PI3K-gamma selective compounds, prioritize the **pyrazine-amine-cyano** pharmacophore identified by the model.
- **Selectivity Optimization:** To avoid cross-reactivity with PI3K-delta, ensure the heterocyclic linker maintains the correct spatial orientation relative to the pyrazine core.
- **Rule 5 Caution:** Chlorinated fused systems appear in many compounds but are not sufficient for gamma selectivity alone — they should be combined with the key pyrazine pharmacophore.

✅ Conclusion

The Balanced Random Forest model successfully learned the structural determinants of PI3K-gamma selectivity, achieving **89.3% accuracy** and **84.7% balanced accuracy**. MolAnchor analysis identified **pyrazine-based scaffolds with specific substitution patterns** as the most critical molecular features driving gamma selectivity, providing actionable insights for medicinal chemistry optimization.